

# Tirumala Venkata Durga Prasad Pitani

AI / GenAI / Agentic AI Engineer — Python | LangChain | RAG | LLM Agents | OpenAI | Hugging Face  
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## Professional Summary

AI / GenAI / Agentic AI Engineer with 2+ years of experience (1.5+ years specialized) designing and productionizing LLM-powered systems: RAG pipelines, autonomous SQL and tool-use agents, and enterprise GenAI applications. Core competency in Python-driven LLM orchestration using LangChain, OpenAI, and Hugging Face. Proven track record of reducing enterprise query resolution time by 60%, achieving 85% NL-to-SQL accuracy, and shipping GenAI MVPs in under one week.

## Technical Skills

**Core Python:** Python 3.x, Async Python, Pandas, NumPy, Pytest, FastAPI, Flask, Django REST Framework  
**GenAI & LLMs:** LangChain, LangGraph, OpenAI API, Hugging Face Transformers, Prompt Engineering, PEFT/LoRA  
**RAG & Vector Search:** FAISS, ChromaDB, Pinecone, Weaviate, Semantic Chunking, Hybrid Retrieval, Re-ranking  
**Agentic AI:** ReAct Agents, Tool-Use Agents, SQL Agents, Multi-Agent Orchestration, LangGraph Workflows  
**Cloud & MLOps:** Azure ML Studio, Azure OpenAI Service, Azure Fabric, PySpark, Docker, CI/CD, Git

## Professional Experience

### AI / GenAI / Agentic AI Engineer

NTT DATA Business Solutions

Oct 2024 – Present  
Bengaluru, India

- Enterprise RAG Pipeline:** Architected a production-grade, end-to-end RAG system using Python, LangChain, FAISS, and FastAPI for enterprise knowledge retrieval; reduced query resolution time by 60% across 500+ daily user interactions.
- Autonomous SQL Agent:** Built a ReAct-based agentic AI system in Python and LangChain enabling natural-language querying over structured enterprise databases; achieved 85% NL-to-SQL accuracy and eliminated 50% of manual analyst query effort.
- LLM-Powered Insight Generator:** Developed an LLM pipeline that automatically synthesizes structured supply-chain data into executive-level narrative insights, compressing report generation from hours to under 2 minutes.
- Prompt Engineering & Optimization:** Systematically iterated prompt templates and retrieval strategies (hybrid BM25 + dense retrieval, re-ranking) to improve RAG answer faithfulness by 30% on domain-specific benchmarks.
- Deployed GenAI services as containerized FastAPI microservices on Azure, ensuring sub-300 ms P95 API latency with JWT-secured endpoints for enterprise consumption.

### Data Science Intern

Viai Solutions Pvt Ltd

Jul 2024 – Aug 2024  
Hyderabad, India

- Engineered automated Python data-ingestion pipelines (Selenium, BeautifulSoup) that fed structured datasets into downstream NLP and analytics workflows, cutting manual effort by 40%.
- Processed and cleaned 100K+ records with Pandas to build high-quality corpora for text analytics and reporting dashboards, improving downstream analysis accuracy.

### Python Developer

Dhanush Infotech Pvt. Ltd.

Jan 2024 – May 2024  
Hyderabad, India

- Built scalable Django REST microservices with JWT authentication and RBAC for 10K+ daily requests — establishing the API layer subsequently extended to serve GenAI inference endpoints.
- Optimized SQL query patterns and database schemas, reducing API response latency by 20%, directly improving performance for LLM context-retrieval queries.

## Projects

### Multi-Agent Enterprise Knowledge Platform | LangGraph, LangChain, FAISS, OpenAI, FastAPI, Python

2026

- Designed a multi-agent LangGraph workflow with specialized sub-agents for retrieval, SQL querying, and report generation; the orchestrator routes user intents autonomously with 88% task-completion accuracy.
- Implemented semantic chunking, metadata-enriched FAISS indexing, and cross-encoder re-ranking to achieve 40% improvement in retrieval precision vs. naive chunk-and-retrieve baseline.

### AI-Powered Analytics Dashboard | Python, Streamlit, LangChain, OpenAI, Plotly

2025

- Delivered a production-ready GenAI analytics MVP in under one week, integrating a LangChain conversational agent with Streamlit UI and Plotly charts for interactive data exploration.
- Implemented conversation memory with LangChain's ConversationBufferWindowMemory to maintain context across multi-turn analytical queries.

### Industrial Defect Detection with Vision LLM Evaluation | PyTorch, CNN, Hugging Face, Transfer Learning

2023

- Benchmarked CNN and pre-trained vision transformer architectures for defect classification, achieving 94% accuracy; published research in institutional repository.

## Education & Certifications

### B.Tech in Computer Science & Engineering — GPA: 7.71/10.0

Indian Institute of Information Technology (IIIT)

2020 – 2024  
Manipur, India

**Certifications:** Microsoft Azure AI Fundamentals (AI-900) | Databricks Lakehouse Fundamentals | Python for Data Science (IIT Bombay NPTEL)

**Achievements:** JEE Mains Qualified — Top 15% nationally | Published Research: CNN-based Industrial Defect Detection

**Languages:** English (Professional) | Telugu (Native)